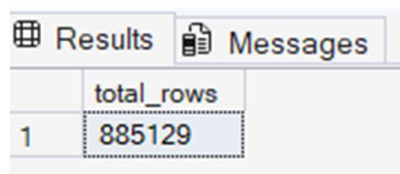


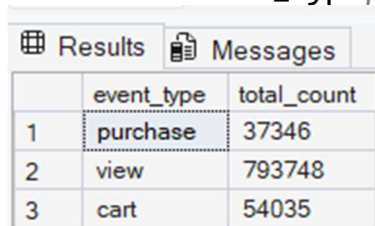
CODE THAT USE IN E-COMMERCE SQL PROJECT

- `SELECT * FROM [e-commerce_data]`
- `USE [E-commerce Database];
SELECT COUNT(*) AS total_rows
FROM [e-commerce_data];`



	total_rows
1	885129

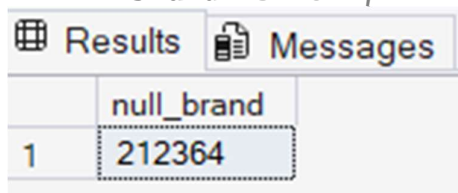
- `USE [E-commerce Database];
SELECT event_type, COUNT(*) AS total_count
FROM [e-commerce_data]
GROUP BY event_type;`



	event_type	total_count
1	purchase	37346
2	view	793748
3	cart	54035

--Data Cleaning and Validation -- NULL Brands

- `USE [E-commerce Database];
SELECT COUNT(*) AS null_brand
FROM [e-commerce_data]
WHERE brand IS NULL;`



	null_brand
1	212364

- **--NULL Category**
`USE [E-commerce Database];
SELECT COUNT(*) AS null_category
FROM [e-commerce_data]
WHERE category_code IS NULL;`

Results Messages	
	null_category
1	236219

- **--NULL Price**

```
USE [E-commerce Database];
SELECT COUNT(*) AS null_price
FROM [e-commerce_data]
WHERE price IS NULL;
```

- **--Price Range**

```
USE [E-commerce Database];
SELECT
ROUND(MIN(price),2) AS min_price,
ROUND(MAX(price),2) AS max_price,
ROUND(AVG(price),2) AS avg_price
FROM [e-commerce_data];
```

Results		Messages	
	min_price	max_price	avg_price
1	0.22	64771.06	146.33

- **--Duplicate Rows**

```
USE [E-commerce Database];
WITH cte AS (
SELECT *,
ROW_NUMBER() OVER (
PARTITION
BY event_time, event_type, product_id, user_id, user_session
ORDER BY event_time
) rn
FROM [e-commerce_data]
)
SELECT COUNT(*) AS duplicate_rows
FROM cte
WHERE rn > 1;
```

Results Messages	
	duplicate_rows
1	655

- **--Replace null brand**

```
USE [E-commerce Database];
UPDATE [e-commerce_data]
SET brand = 'Unknown'
WHERE brand IS NULL;
```

- **--Replace null category**

```
USE [E-commerce Database];
UPDATE [e-commerce_data]
SET category_code = 'Unknown'
WHERE category_code IS NULL;
```
- **--Remove Duplicate rows**

```
USE [E-commerce Database];
WITH cte AS (
SELECT *,
ROW_NUMBER() OVER (
PARTITION BY
event_time,event_type,product_id,user_id,user_session
ORDER BY event_time
) rn
FROM [e-commerce_data]
)
DELETE FROM cte
WHERE rn > 1;
```
- **--BUSINESS UNDERSTANDING**

--Event Distribution

```
SELECT event_type, COUNT(*) AS total_events
FROM [e-commerce_data]
GROUP BY event_type
ORDER BY total_events DESC;
```

Results Messages		
	event_type	total_events
1	view	793099
2	cart	54032
3	purchase	37343

- **--Percentage distribution**

```
SELECT
event_type,
COUNT(*) AS total_events,
CAST(ROUND(100.0 * COUNT(*) / SUM(COUNT(*)) OVER(),2) AS
decimal(10,2)) AS percentage_share
FROM [e-commerce_data]
GROUP BY event_type;
```

Results Messages			
	event_type	total_events	percentage_share
1	purchase	37343	4.22
2	view	793099	89.67
3	cart	54032	6.11

- --Final Insights

```
SELECT
SUM(CASE WHEN event_type = 'view' THEN 1 ELSE 0 END) AS
views,
SUM(CASE WHEN event_type = 'cart' THEN 1 ELSE 0 END) AS
carts,
SUM(CASE WHEN event_type = 'purchase' THEN 1 ELSE 0 END) AS
purchases
FROM [e-commerce_data];
```

Results		Messages	
	views	carts	purchases
1	793099	54032	37343

- --Total Revenue

```
SELECT ROUND(SUM(price),2) AS total_revenue
FROM [e-commerce_data]
WHERE event_type = 'purchase';
```

Results		Messages	
	total_revenue		
1	5125113.92		

- --Total Purchases

```
SELECT COUNT(*) AS total_purchases
FROM [e-commerce_data]
WHERE event_type = 'purchase';
```

Results		Messages	
	total_purchases		
1	37343		

- --Average Order Value

```
SELECT ROUND(AVG(price),2) AS avg_order_value
FROM [e-commerce_data]
WHERE event_type = 'purchase';
```

Results		Messages	
	avg_order_value		
1	137.24		

- --Revenue by Brand

```
SELECT TOP 10
brand,
ROUND(SUM(price),2) AS revenue
FROM [e-commerce_data]
WHERE event_type='purchase'
AND brand <> 'Unknown'
GROUP BY brand
ORDER BY revenue DESC;
```

	brand	revenue
1	msi	643492.34
2	gigabyte	556183.03
3	palit	484210.57
4	asus	330147.11
5	sapphire	306192.94
6	amd	191987.4
7	canon	137964.79
8	powercolor	98804.38
9	epson	98262.95

- --Revenue by Category

```
SELECT TOP 10
category_code,
ROUND(SUM(price),2) AS revenue
FROM [e-commerce_data]
WHERE event_type='purchase'
AND category_code <> 'Unknown'
GROUP BY category_code
ORDER BY revenue DESC;
```

	category_code	revenue
1	computers.components.videocards	2604764.8
2	computers.peripherals.printer	364566.8
3	computers.components.cpu	219168.98
4	electronics.telephone	150915.69
5	stationery.cartrige	103595.99
6	computers.components.motherboard	102871.16
7	electronics.audio.acoustic	84724.8
8	auto.accessories.player	81558.66
9	computers.notebook	63604.79

- --Funnel Breakdown

```
SELECT event_type, COUNT(*) AS users
FROM [e-commerce_data]
GROUP BY event_type;
```

	event_type	users
1	purchase	37343
2	view	793099
3	cart	54032

- --Conversion Rate (View → Purchase)

```
SELECT
CAST(100.0 * SUM(CASE WHEN event_type='purchase' THEN 1
ELSE 0 END) /
NULLIF(SUM(CASE WHEN event_type='view' THEN 1 ELSE 0
END),0)
AS decimal(10,2)) AS conversion_rate
FROM [e-commerce_data];
```

Results Messages	
	conversion_rate
1	4.71

- --Cart Abandonment Insight

```
SELECT
CAST(100.0 * SUM(CASE WHEN event_type='cart' THEN 1 ELSE 0
END) /
NULLIF(SUM(CASE WHEN event_type='view' THEN 1 ELSE 0
END),0)
AS decimal (10,2)) AS view_to_cart_rate
FROM [e-commerce_data];
```

Results Messages	
	view_to_cart_rate
1	6.81

- --Drop-off Analysis

```
SELECT
SUM(CASE WHEN event_type='view' THEN 1 ELSE 0 END) AS
views,
SUM(CASE WHEN event_type='cart' THEN 1 ELSE 0 END) AS
carts,
SUM(CASE WHEN event_type='purchase' THEN 1 ELSE 0 END) AS
purchases,
(
SUM(CASE WHEN event_type='view' THEN 1 ELSE 0 END)
- SUM(CASE WHEN event_type='purchase' THEN 1 ELSE 0 END)
) AS drop_off
FROM [e-commerce_data];
```

Results Messages				
	views	carts	purchases	drop_off
1	793099	54032	37343	755756

- --Top product by revenue

```
SELECT TOP 10
product_id,
ROUND(SUM(price),2) AS revenue
FROM [e-commerce_data]
WHERE event_type='purchase'
GROUP BY product_id
ORDER BY revenue DESC;
```

	product_id	revenue
1	1821813	213844.25
2	4099645	165156.11
3	3791351	86414.67
4	4079420	85856.41
5	1830099	82270.89
6	893196	82214.4
7	4078916	81445.84
8	4102739	78313.06
9	4079565	69621.04

- --Brand Contribution %

```
SELECT
brand,
ROUND(SUM(price),2) AS revenue,
ROUND(100.0 * SUM(price) / SUM(SUM(price)) OVER(),2) AS
revenue_share
FROM [e-commerce_data]
WHERE event_type='purchase'
AND brand <> 'Unknown'
GROUP BY brand;
```

	brand	revenue	revenue_share
1	lucia	295.41	0.01
2	redline	91.14	0
3	viewsonic	5032.88	0.11
4	kada	3066.37	0.07
5	merrylock	2804.92	0.06
6	wacom	2622.47	0.06
7	sony	35051.28	0.77
8	peakmeter	188.24	0
9	acv	9649.66	0.21

- --Average Price by Category

```
SELECT
category_code,
ROUND(AVG(price),2) AS avg_price
FROM [e-commerce_data]
WHERE event_type='purchase'
GROUP BY category_code
ORDER BY avg_price DESC;
```

Results		Messages
	category_code	avg_price
1	electronics.video.projector	562.03
2	computers.components.videocards	378.16
3	country_yard.cultivator	351.56
4	sport.bicycle	284.4
5	appliances.kitchen.meat_grinder	282.64
6	electronics.camera.video	272.72
7	computers.peripherals.monitor	229.73
8	construction.tools.saw	214.64
9	appliances.sewing_machine	212.51

- --Active Users (Unique Users)
SELECT COUNT(DISTINCT user_id) AS total_users
FROM [e-commerce_data];

Results		Messages
	total_users	
1	407283	

- --Session Activity
SELECT COUNT(DISTINCT user_session) AS total_sessions
FROM [e-commerce_data];

Results		Messages
	total_sessions	
1	490398	

- --Avg Events per User (Engagement)
SELECT
CAST(ROUND(COUNT(*) * 1.0 / COUNT(DISTINCT user_id),2) AS
decimal(10,2)) avg_events_per_user
FROM [e-commerce_data];

Results		Messages
	avg_events_per_user	
1	2.17	

- --Most Active Users (Top customers by activity)

```
SELECT TOP 10
user_id,
COUNT(*) AS total_actions
FROM [e-commerce_data]
GROUP BY user_id
ORDER BY total_actions DESC;
```

	user_id	total_actions
1	1515915625554995474	572
2	1515915625527763086	424
3	1515915625591251010	363
4	1515915625591659523	339
5	1515915625537803839	329
6	1515915625568273951	312
7	1515915625599852988	288
8	1515915625598794428	260
9	1515915625536567608	259

- --Most Purchased Users

```
SELECT TOP 10
user_id,
COUNT(*) AS total_purchases
FROM [e-commerce_data]
WHERE event_type='purchase'
GROUP BY user_id
ORDER BY total_purchases DESC;
```

	user_id	total_purchases
1	1515915625601579158	56
2	1515915625591659523	49
3	1515915625596534847	45
4	1515915625604175669	43
5	1515915625605054644	42
6	1515915625599634258	39
7	1515915625521745364	36
8	1515915625603840072	33
9	1515915625603113992	28

- --Monthly revenue

```
SELECT
FORMAT(CAST(REPLACE(event_time, ' UTC', '') AS DATETIME),
'yyyy-MM') AS month,
ROUND(SUM(price),2) AS revenue
FROM [e-commerce_data]
WHERE event_type = 'purchase'
GROUP BY FORMAT(CAST(REPLACE(event_time, ' UTC', '') AS
DATETIME), 'yyyy-MM')
ORDER BY month;
```

	month	revenue
1	2020-09	96353.01
2	2020-10	562590.39
3	2020-11	787886.24
4	2020-12	813068.01
5	2021-01	1488410.57
6	2021-02	1376805.7

- --Monthly Purchases

```
SELECT
FORMAT(CAST(REPLACE(event_time, ' UTC', '') AS DATETIME),
'yyyy-MM') AS month,
COUNT(*) AS total_purchases
FROM [e-commerce_data]
WHERE event_type = 'purchase'
GROUP BY FORMAT(CAST(REPLACE(event_time, ' UTC', '') AS
DATETIME), 'yyyy-MM')
ORDER BY month;
```

	month	total_purchases
1	2020-09	1007
2	2020-10	6276
3	2020-11	7574
4	2020-12	6529
5	2021-01	8315
6	2021-02	7642